

used to denote actual transaction volume, transaction value (in dollars), as well as number of contracts, etc. Shadow liquidity refers to the underlying stock liquidity for a financial product where the underlying pricing scheme is a financial instrument that trades on its own in the market. The term is most commonly used to refer to the underlying stock volume for an ETF instrument. For example, investors wishing to buy an ETF can either purchase the ETF in the market or purchase the underlying stocks that comprise the ETF in the market and then create the ETF. Hence, investors wishing to purchase an ETF have two available sources of liquidity that can be used to complete the transaction. Mispricing liquidity refers to the volume that is on-standby in the market and ready to transact if there is a mispricing between two instruments or an arbitrage opportunity. The most common occurrence of mispricing liquidity is associated with statistical arbitrage traders who are standing by ready to transact in an index and its underlying stock members or in an exchange traded fund and its underlying stock members if there is a market mispricing. These traders will sell (short) shares in the over-valued instrument and buy shares in the under-valued instrument. Factor exposure liquidity refers to the investors' ability to invest in a similar financial instrument which provides the risk characteristics and expected returns stream as the desired instrument. For example, investors interested in gaining exposure to the SP500 index have numerous options available. They can purchase the stocks that comprise the SP500 index, purchase any of the large cap SP500 index exchange traded funds, an SP500 futures contract, a mini futures contract, etc. All of these instruments will provide the investor with exact same returns and risk. They all have the same risk composition and same stream of future returns.

Competition refers to the investors' ability to transact the financial instrument from various venues, broker-dealers, and/or market participants. Competition has been found to dramatically reduce trading costs in the equity markets. A marketplace with multiple venues is more competitive and cost efficient than a market with a single or relatively few dealers for the financial instrument.

Observations

The following results are based on empirical data and market observations over the period 1H2011 and trade costs estimates were based on a transaction size of \$10 million USD.

Equities. A trade value of \$10 million USD is equivalent to an order size of 4% ADV for a US large cap stock and an order size of 80% ADV for a US small cap stock. The reason the small cap order size is so much